

KOLO, Poland

WMO: 123450

Lat: 52.1998N

Lon: 18.6602E

Elev: 384

StdP: 14.49

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	13.1	2.3	6.3	8.8	8.0	8.4	14.2	27.6	37.7	25.3	34.4	7.5	60	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	88.6	73.0	86.0	71.3	83.0	69.2	76.0	84.4	74.5	81.6	72.4	79.0	9.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	126.4	80.4	72.1	120.5	79.6	69.9	111.7	77.9	39.7	85.0	38.2	81.6	36.3	78.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.7	19.1	17.1	DB	3.4	93.4	7.9	3.0	-2.4	95.5	-7.0	97.3	-11.4	98.9	-17.2	101.1	(4)
			WB	2.8	74.0	7.8	2.4	-2.8	75.7	-7.4	77.1	-11.7	78.4	-17.4	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	32.7	35.1	39.3	50.2	58.3	63.5	67.7	66.9	59.1	51.0	41.1	35.1
	DBStd	15.69	13.60	12.83	9.52	9.80	8.17	6.59	6.51	5.84	7.43	9.50	9.37	11.52
	HDD50	2401	575	454	355	110	16	0	0	6	96	297	492	
	HDD65	5881	1012	846	801	463	241	105	44	47	210	454	725	933
	CDD50	2430	38	37	23	116	273	405	550	523	278	127	29	31
	CDD65	435	10	10	5	18	33	60	128	105	32	20	7	7
	CDH74	4004	62	60	43	160	351	626	1197	890	323	189	56	47
	CDH80	1215	14	13	14	43	88	193	405	264	100	59	12	10
Wind	WSAvg	8.5	9.6	9.3	9.8	9.1	8.1	7.7	7.6	7.2	7.9	8.3	8.7	9.1
Precipitation	PrecAvg	20.8	1.1	1.0	1.4	1.2	2.2	2.2	3.3	2.2	1.9	1.4	1.5	1.3
	PrecMax	28.2	2.9	2.0	3.0	3.8	6.1	5.9	7.7	7.5	4.1	4.0	5.2	3.7
	PrecMin	13.5	0.1	0.2	0.6	0.2	0.3	0.5	1.3	0.6	0.3	0.1	0.0	0.2
	PrecStd	3.5	0.7	0.5	0.6	0.8	1.2	1.3	1.6	1.4	1.1	0.9	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.6	82.3	79.6	86.7	87.5	91.3	91.8	90.5	87.9	87.2	82.6	80.8
		MCWB	69.5	68.3	66.9	70.1	72.2	71.7	72.6	73.5	75.5	76.4	73.1	70.9
	2%	DB	74.6	75.2	67.3	79.7	82.9	86.3	88.2	86.7	83.4	81.6	72.5	72.5
		MCWB	66.8	66.5	61.4	66.7	68.1	71.0	72.3	71.6	71.6	74.2	68.7	66.8
	5%	DB	67.8	68.1	57.8	73.0	78.2	81.3	84.9	82.9	77.4	74.9	57.0	61.4
		MCWB	63.6	63.8	49.4	63.7	65.6	67.2	69.6	68.2	67.8	71.1	53.1	57.6
	10%	DB	49.9	53.2	51.7	67.1	73.6	77.2	80.9	79.2	73.1	66.5	51.5	47.6
		MCWB	46.1	48.7	45.8	57.5	63.8	65.1	67.5	66.4	64.7	60.0	48.9	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.5	70.2	67.9	72.0	74.4	76.0	77.0	77.1	76.8	76.7	73.0	71.6
		MCDB	78.9	77.2	77.8	80.7	83.2	85.8	86.7	86.9	85.0	84.6	80.5	78.5
	2%	WB	67.8	67.6	61.1	69.1	71.8	73.5	75.4	75.0	74.8	74.5	67.7	67.2
		MCDB	73.7	73.6	67.9	77.6	79.8	81.8	83.6	82.1	80.9	80.8	72.3	70.8
	5%	WB	62.6	63.4	49.9	64.7	68.3	69.4	73.0	71.7	71.3	71.2	53.9	56.6
		MCDB	67.5	69.7	56.0	73.1	75.5	78.7	80.5	78.7	74.1	75.4	56.5	61.1
	10%	WB	46.3	48.5	46.3	56.7	63.6	66.2	69.3	68.1	63.0	59.6	49.0	45.5
		MCDB	48.7	51.5	50.5	66.1	73.5	75.8	79.3	77.3	71.0	64.7	51.2	47.4
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.7	13.1	17.0	18.7	18.7	18.2	18.4	16.5	13.7	9.3	8.3
		MCDBR	25.3	20.6	22.6	23.5	25.1	26.5	23.6	24.3	24.4	23.4	18.3	21.8
		MCWBR	15.2	9.2	13.0	12.1	12.7	12.6	9.9	10.9	12.2	13.1	11.7	11.5
	5% WB	MCDBR	23.3	19.1	20.0	22.1	24.1	25.0	22.1	23.1	23.5	21.4	17.4	21.3
		MCWBR	13.5	8.6	12.6	11.8	14.0	13.8	11.2	12.3	15.1	13.4	12.2	11.5
Clear-Sky Solar Irradiance	taub	0.315	0.342	0.377	0.425	0.419	0.412	0.444	0.431	0.393	0.368	0.347	0.312	
	taud	2.271	2.271	2.225	2.146	2.219	2.268	2.196	2.234	2.312	2.355	2.329	2.304	
	Ebn at Noon	212	241	255	256	264	267	255	252	248	229	198	192	
	Edh at Noon	21	28	35	43	42	41	43	39	32	26	20	17	
All-Sky Solar Radiation	RadAvg	216	432	828	1316	1646	1720	1596	1416	1012	563	246	161	
	RadStd	40	74	104	172	170	160	187	119	141	85	31	21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.97	N/A	N/A	N/A	N/A	N/A	N/A	-385	+191	+70
Regional (21 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-306	+148	+50

Nomenclature: See separate page